Nifty Volatility Forecaster

Project summary. State as of the evening of Tuesday 3 November 2026, India time.

- **The project works from raw data to a working app.** Tonight it produced a real forecast for Wednesday 4 November 2026, from a model frozen long before that date. The Transformer has now also been compared with GARCH, an MLP and an LSTM.
- **It isn't a full daily pipeline yet.** Someone has to run a command each evening, the prices come from an unofficial source, and nothing records how live forecasts turn out.
- **I need three things from you to start closing that:** a yes to a scheduled task, a yes to keeping a forward record, and a newer minute-data download. Section 2 has these and the decisions that can wait.
- **The results are useful but limited.** The Transformer's squared error is about 9 to 10% below the HAR benchmark's, but it didn't significantly beat the simpler MLP or LSTM, so its extra complexity isn't shown to pay off. News added nothing measurable, and its 99% VaR runs too low to use as a limit. Section 3 explains each result.

Forecast for Wed 4 Nov 2026

±0.60%

One standard deviation of the session's move, about 142 points from the 23,810.85 close.

Likely range for actual volatility, about 80% of the time	±0.43% to ±0.88%
Annualised volatility	9.4%
99% VaR (runs too low, see section 3)	1.39% loss
HAR benchmark	±0.59%
India VIX implied, includes overnight	±0.99%
What actually happened on 3 Nov	±0.54%

Made at 19:42 IST from 60 sessions of 5-minute bars, India VIX 15.72 and turnover 32% above its 20-day average. It has no outcome yet and is in no score.

1. Where the project stands

Part	State
Data	Nifty minute bars from 2015 to 15 May 2026, India VIX and turnover from NSE to 2 Nov 2026, and news headlines to 30 Jun 2023.
Models	Two Transformers written in NumPy, with and without news, three seeds each. Trained on data to the end of 2019, validated on 2020, then frozen with recorded checksums. GARCH(1,1), an MLP and an LSTM were added for comparison, also written by hand and frozen before their test forecasts were made. The MLP and LSTM use the news features.
Test results	615 forecasts made from 1 Jan 2021 to 30 Jun 2023, scored once against the HAR, naive and VIX benchmarks. GARCH, the MLP and the LSTM were later scored once on the same days, with no Transformer number changing.
Later check	705 further no-news forecasts made from 3 Jul 2023 to 13 May 2026, kept separate from the test results. The comparison models aren't in it.
Live forecast	`python -m src.live` forecasts the next session with the frozen no-news model. It stops without writing anything if an input is missing or the sources disagree. It also works out HAR, naive and VIX for reference. GARCH, the MLP and the LSTM don't run live.
App	The page opens on the live forecast. Below it are a replay of 1,320 past days, forecast charts, a risk desk, a scoreboard of all 8 models and model details. The backend only reads saved results. It isn't running at the moment, so it has to be started before the page will load data.

Part	State
Checks	All 336 backend and 106 frontend tests pass, and browser checks pass on the dev server. An audit confirmed that adding the comparison left every existing forecast, checkpoint and score unchanged. The production build compiles but hasn't been click-tested since the live section was added. Run on the training feed, the live code matched 60 published later-check forecasts to rounding error.

What's still missing

- **Nothing runs by itself.** The forecast goes out of date when its session closes at 15:30 unless someone runs the command that evening, and a failed run alerts no one.
- **No live track record.** Forecasts aren't saved and compared with what happened next, so tonight's number has no live evidence behind it.
- **The comparison models can't run live as they stand.** The MLP and LSTM were trained with news features, and the news archive ends in June 2023. GARCH needs an unbroken run of daily returns, which has the gap described below, though NSE's daily files could probably fill it. None of the three has been scored after June 2023 either.
- **The price feed is unofficial and only partly checked.** Yahoo's bars agree with NSE's daily figures, but they share no dates with the training feed, so the two haven't been compared directly. The one difference I found, Yahoo's last bar usually being the official close, moved past forecasts by a median of 0.003 percentage points and at most 0.08. I haven't checked Yahoo's or NSE's terms of use.
- **A gap in history.** There are no forecasts between 14 May and 4 November 2026, because the minute file stops in May and Yahoo only reaches back about 60 days.
- **Old training data.** The models learned from 2015 to 2019. They've held up against HAR since, but nothing has refreshed them.
- **Smaller items.** NSE's holiday list only covers the current year, so the turn of the year needs handling. Special weekend sessions aren't predicted, and the weekly expiry rule would need updating if NSE changes it. The masthead and ticker still follow the replay day. The project isn't in version control.

What "real time" can mean for this model

The model reads a complete trading day, so it makes one forecast each evening for the next session. It can't update during the session; that would be a different model. I've taken your goal to mean this: **every evening, without you doing anything, the app shows a forecast for the next session built from that day's data, and every past forecast is scored against what happened.** Tell me if you meant intraday updates, because that's a much larger piece of work.

2. What I need from you

To get started

#	What I need	Why
1	**A yes to a scheduled task on this PC**	It would run on weekday evenings at about 19:00 IST and retry if NSE's file is late. This is what ends the manual step. The PC has to be on at that time. I haven't set it up because it runs outside the project without you starting it.
2	**A yes to keeping a forward record**	Each forecast is saved before its outcome exists, then scored the next evening and shown apart from the test and later-check scores. It's the only real evidence of how the model does from here, and every day without it is lost.
3	**A newer download of the Kaggle Nifty 50 minute dataset (debashis74017)**	It lets me compare Yahoo's bars with the training feed directly and fill the gap after May, as far as the file reaches. It needs to reach dates inside Yahoo's last 60 days, so download it close to when you send it. Put it in `data/live/` and leave the file in `data/raw/` alone, since replacing that would change the study's inputs.

Decisions that can wait

#	Decision	What it affects	My recommendation
4	**Who is this for: you alone, others, or something commercial?**	Whether free data is acceptable, and whether hosting and legal questions apply.	Treat it as personal for now. Before sharing it widely or selling it, read the data terms and get advice on SEBI rules, which I can't assess.
5	**Keep Yahoo, or pay for a broker data API?**	Reliability. A broker API needs your account and API key, which you'd keep yourself. Kite Connect was ₹500 a month when I checked, and Upstox has minute data from January 2022.	Stay on Yahoo while the forward record builds. Switch if it breaks or others start relying on this. First check whether the API needs a daily login, which would get in the way of unattended runs.
6	**Run on this PC, or on an always-on machine?**	Missed evenings when the PC is off. A hosted machine needs an account and costs money.	Start on this PC. Move if evenings get missed.
7	**How should a failed run reach you?**	Whether you notice a failure the same evening. Email would need a sending account from you.	Start with a Windows notification, plus the page's out-of-date banner.
8	**Keep the models frozen, or retrain them on a schedule?**	The models are 2019-vintage, and your rule so far is no retraining. A retrained model would need new, unseen days to be judged on.	Stay frozen for about a year of forward record, then decide from the evidence. Any retrained model should run alongside the frozen one and be scored separately.
9	**Should simpler models run alongside the live one?**	The no-news Transformer is the only neural network that can run live, and the test period didn't show it beating the simpler networks. Without them, the live record can't say whether its complexity is worth keeping.	Yes. Train no-news versions of the MLP and LSTM, which takes about a minute and changes no reported result. Score them and GARCH on the later check, then on the forward record, kept apart. Keep the Transformer as the headline unless that record says otherwise.
10	**Keep the VaR rule as it is, or add a calibrated figure?**	Today's rule breaches about four times too often, so it can't serve as a limit.	Keep the rule with its warning, and add a calibrated figure judged only on new days. GARCH's line, the only pass on the test period, is one candidate.
11	**Bring news back?**	It would need a current feed of the same kind of headlines, likely paid.	Leave it out. It added nothing measurable in the study.
12	**Put the app online?**	Only matters if others need to see it. It would need hosting, and possibly a domain, from you.	Not yet. Follows from decision 4.
13	**Put the project in git?**	There's no history to return to if an unattended change goes wrong. I'd only commit when you ask.	Yes, before automating anything.

What I'd build once you've said yes

None of this needs anything further from you:

- The evening runner, with retries and a log of each run.
- The forward record and a page section showing forecast against actual, with running scores against HAR. I can start it with about 17 recent sessions that Yahoo's history already covers, clearly marked as backfilled.
- The direct comparison of Yahoo's bars with the training feed, using your download.
- Failure alerts, handling for the turn of the year, and clean-up of old downloads.

- A masthead and ticker that follow the live day, plus a full click-test of the production build.

It's done when: a forecast appears each evening with nobody touching it, a failed run reaches you the same evening, the page never shows an old forecast without saying so, and every forecast gets scored the next evening.

3. What the results mean

How to read the numbers

- **Realized variance (RV)** adds up the squared 5-minute returns within one session. The models forecast its logarithm, and the app turns that into a "±%" move. It covers the session only, so overnight gaps aren't in it.
- **MSE** is the average squared error in log RV. **QLIKE** is a second error measure that punishes under-forecasting volatility more heavily. Lower is better for both.
- **Diebold-Mariano (DM)** asks whether the gap between two models is bigger than chance would produce. A negative statistic means the first model had the lower error. Running many comparisons makes a chance result more likely, so each group of tests gets a strict Bonferroni cut-off for p, shown in the table below. More tests mean a stricter cut-off.
- **99% VaR** is a loss level that should be passed on only about 1% of days. A **breach** is a day that passed it. **Kupiec** tests whether the breach count fits 1%, and it fails above 3.84.
- **HAR** is the standard simple benchmark, a regression on daily, weekly and monthly volatility. **Naive** just repeats today's volatility.
- **GARCH(1,1)** is the classic volatility model, with 3 parameters. It updates tomorrow's variance from today's squared return and today's variance. The **MLP** is a plain neural network that sees the last 5 days at once. The **LSTM** is a neural network that reads the last 22 days in order.

Test period: 615 days, 6.15 VaR breaches expected. Simplest model first

Model	MSE	QLIKE	VaR breaches	Kupiec LR	VaR test
Naive (repeat today)	0.3609	0.2042	26	35.92	Failed
India VIX implied	1.6748	0.5352	0	12.36	Failed
HAR benchmark	0.2705	0.1571	26	35.92	Failed
GARCH(1,1)	0.5443	0.2144	8	0.51	Passed
MLP	0.2536	0.1438	25	33.01	Failed
LSTM	0.2485	0.1434	24	30.19	Failed
Transformer, no news	0.2453	0.1437	28	41.98	Failed
Transformer, with news	0.2463	0.1338	24	30.19	Failed

Later check: 705 days, 7.05 VaR breaches expected

Model	MSE	QLIKE	VaR breaches	Kupiec LR	VaR test
Transformer, no news	0.3559	0.2619	24	25.32	Failed
HAR benchmark	0.3953	0.2834	28	35.97	Failed
Naive (repeat today)	0.5656	0.3885	34	54.14	Failed
India VIX implied	1.5451	0.4999	6	0.17	Passed

Size, training time, and each seed's own score (mean ± spread)

Model	Parameters	Training	MSE by seed	QLIKE by seed
HAR benchmark	4	under 0.001s	n/a	n/a
GARCH(1,1)	3	0.3s	n/a	n/a
MLP	7,937	0.7s	0.2782 ± 0.0122	0.1577 ± 0.0124
LSTM	6,561	45.4s	0.2581 ± 0.0108	0.1493 ± 0.0146
Transformer, no news	9,665	31.2s	0.2654 ± 0.0220	0.1562 ± 0.0188
Transformer, with news	9,761	37.2s	0.2841 ± 0.0236	0.1560 ± 0.0133

Is the gap real? Diebold-Mariano tests

Comparison	MSE statistic	MSE p	QLIKE statistic	QLIKE p	Strict cut-off
Test: with news vs naive	-5.89	<0.0001	-6.31	<0.0001	0.0036
Test: with news vs VIX	-18.55	<0.0001	-15.91	<0.0001	0.0036
Test: with news vs GARCH	-9.38	<0.0001	-6.08	<0.0001	0.0036
Test: with news vs HAR	-2.24	0.0250	-2.89	0.0038	0.0036
Test: with news vs MLP	-0.70	0.4861	-1.44	0.1503	0.0036
Test: with news vs LSTM	-0.26	0.7959	-1.84	0.0657	0.0036
Test: with news vs no news	0.12	0.9010	-1.91	0.0558	0.0036
Test: no news vs HAR	-2.80	0.0052	-1.97	0.0487	0.0042
Test: no news vs MLP	-0.78	0.4353	-0.01	0.9942	0.0042
Test: no news vs LSTM	-0.41	0.6804	0.07	0.9469	0.0042
Later: no news vs HAR	-3.56	0.0004	-1.51	0.1310	0.0125
Later: no news vs naive	-8.82	<0.0001	-6.07	<0.0001	0.0125

A negative statistic means the first model had the lower error. The MLP, LSTM and GARCH weren't tested against HAR, only against the Transformers.

The model clearly beats the simplest approaches

Its MSE is 32% below naive on the test period and 37% below on the later check. On the test period its leads over naive, India VIX and GARCH all pass the strict cut-off on both scores.

It doesn't clearly beat the simpler neural networks

The with-news Transformer ranks 2nd of 8 on MSE, behind the no-news Transformer, and 1st on QLIKE. But none of its leads over the MLP, the LSTM or the no-news model is statistically significant, even before any adjustment. The MLP trains in under a second and its MSE is only 3% higher. Scored one seed at a time, the LSTM has the lowest mean on both scores: MSE 0.2581 against the Transformer's 0.2841, and QLIKE 0.1493 against 0.1560.

So the Transformer's extra complexity isn't shown to pay off. That doesn't make the LSTM the better model: its leads weren't shown to be real either, three seeds is few, and only the Transformer's settings were chosen on validation data. The live model, the no-news Transformer, is just as hard to separate from them (p from 0.44 to 0.99).

It beats HAR, but modestly, and the evidence is mixed

MSE is about 9% lower on the test period and 10% lower on the later check. With 14 tests in the group, neither test-period gap passes the strict cut-off of 0.0036. QLIKE just misses at p = 0.0038, though it passed the earlier cut-off of 0.0083, set when only six tests had been run, and MSE is further off at p = 0.0250. The no-news model misses too (p = 0.0052 and 0.0487). On the later check MSE is strong (p = 0.0004) and QLIKE isn't significant (p = 0.1310).

The test-period QLIKE lead also comes mainly from a few sessions. The Transformer is ahead of HAR on only 50% of days, and its 10 best days supply 59% of the lead. It's ahead in 72% of rolling 63-day windows. The fair reading is a small edge that probably exists but isn't firmly proven. On the 4 Nov forecast the two differ by only 0.01 percentage points.

The edge over HAR shows most in the more volatile stretches

Splitting test days into three groups by the 22-day volatility known in advance, the QLIKE lead over HAR is clear in the most volatile third (p = 0.0003) and not significant in the calm one (p = 0.2766), where the MLP has the best QLIKE. These groups were split after the fact, so treat this as description. "Most volatile" also only reaches 15% annualised, because the 2020 crash fell in the validation period, so it says little about a crisis.

GARCH passes the VaR test but forecasts RV poorly

It's the only model to pass Kupiec on the test period, with 8 breaches against about 6 expected, yet its MSE of 0.5443 is worse than naive's. Both come from the same fact: GARCH models the variance of the open-to-close return, which runs above RV. Its forecasts sit 0.472 log units above log RV on average, which keeps its VaR line further away but counts as error when scored on RV. So its pass says its VaR scale is right, not that it's the better forecaster.

Forecasts are useful, not precise

An MSE of 0.25 means the actual move typically lands within a factor of about 1.28 of the forecast, and the later check's 0.36 means about 1.35. That's why a ±0.60% forecast comes with a range of ±0.43% to ±0.88%. The model estimates how large moves are likely to be. It says nothing about direction.

Errors grew after the test period, for the simple benchmarks too

The model's MSE rose from about 0.25 to 0.36. HAR's also rose, from 0.27 to 0.40, and naive's from 0.36 to 0.57, and the model kept its lead. That points to a harder period more than to this model aging badly, though it doesn't rule out aging.

Averaging the three seeds matters

On the test period only three of the six single Transformers beat HAR's 0.2705, and their scores ranged from 0.244 to 0.310. Averaging helps the MLP and LSTM too, taking their mean seed MSE of 0.2782 and 0.2581 down to 0.2536 and 0.2485. All three no-news seeds beat HAR on the later check, but a single trained copy still shouldn't be trusted alone. The live forecast uses the average.

News added nothing measurable

The MSE difference between the with-news and no-news models is effectively nil (p = 0.90), and the QLIKE gain just misses 5% (p = 0.056). That doesn't prove news is useless, only that this news signal didn't help. The sentiment classifier scored 0.806 accuracy and 0.748 macro F1 on its financial-sentence holdout, but its accuracy on Indian headlines was never measured. Losing news therefore costs the live forecast little.

The 99% VaR is not safe to use as a limit

The with-news model breached 24 times against about 6 expected on the test period, and the no-news model 24 against about 7 on the later check. That's 3 to 4% of days rather than 1%, and every model built to forecast RV failed Kupiec in every period it was scored on. So the 1.39% figure for 4 Nov is better read as a loss passed on about 3 to 4% of days, going by past form.

The fault lies mainly in the scale, not the forecasts. Squared open-to-close returns averaged 1.31 times RV on the test period and 1.19 times on the later check, partly because RV misses the move from the open to the first 5-minute close. Given the true RV, the same rule breaches 6 times on the test period, which is on target, and 10 on the later check, which is close.

India VIX measures something different

Its implied volatility runs high by about 1.17 log points on the test period, or about 1.8 times the session's move, because it includes overnight moves and a risk premium. That's why it scores worst on MSE and why its VaR errs on the cautious side: 0 breaches on the test period, which also fails Kupiec, and 6 on the later check, which passes. A pass from a measure biased this high is not evidence that it sizes risk accurately.

Attention doesn't explain the forecast

Average attention weights sit between about 0.04 and 0.05, against 0.045 if they were perfectly even. They don't show which past days drove a forecast, so the heatmap shouldn't be read as an explanation.

Choices that affect how far to trust the scores

Validation MSE was 0.3888 with news and 0.4013 without, against 0.4118 for HAR and 0.4219 for naive. Those are slightly flattering, because last-day pooling replaced mean pooling after mean pooling scored poorly there (0.73 and 0.76). The test set hadn't been scored at that point. An early data audit printed statistics that included test rows, though nothing was tuned from them. Expiry flags before 2019 are a project choice and not all real expiries.

The comparison isn't a level contest. The Transformer's training settings were chosen on validation data, while the MLP and LSTM use the settings you specified, untuned, so they may be capable of better. The Transformer's test scores were also already known when the new models were added, though nothing about them was chosen after seeing test results. They use the project's 18 features rather than the brief's 17, so the MLP has 90 inputs rather than 85.

The bottom line

Use the forecast as a reasonable estimate of how much the Nifty may move during the next session, somewhat better than a good simple benchmark. It hasn't been shown to beat simpler neural networks, so its value lies in the forecast, not in the Transformer as such. Don't use it to predict direction, don't use its VaR as a limit, and don't treat it as trading advice. All of this evidence comes from past data. Whether it holds up live is still unknown, and the forward record is what would answer that.